# What Do Tabulated Isotherms Certify? Sharp Certain Answers to Mixture-Adsorption Screening Queries

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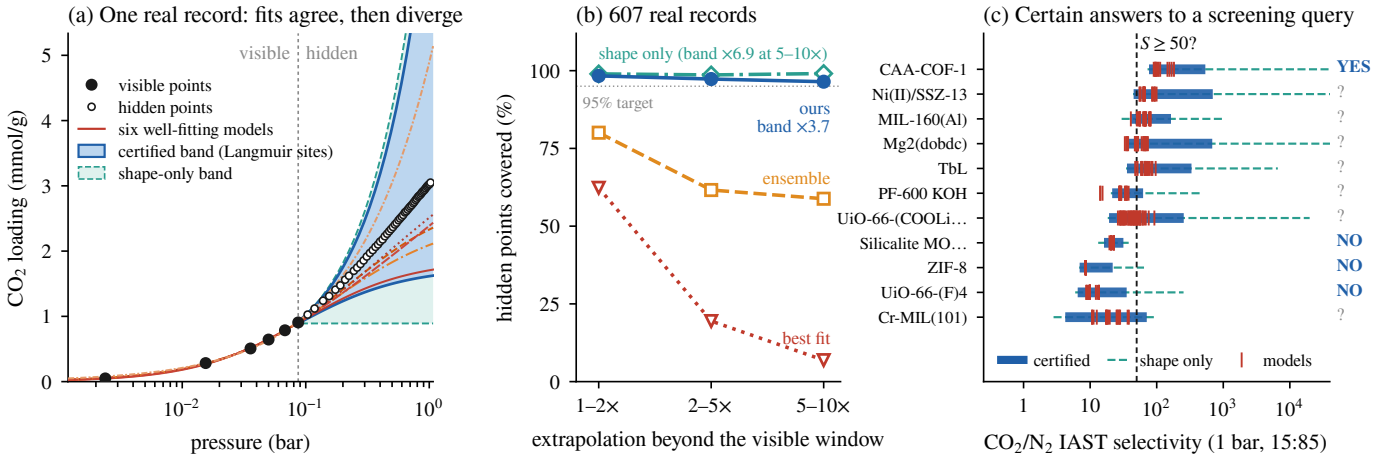



Fig. 1. Fits that agree on the measured window diverge beyond it; a certified band covers what was hidden and turns a screening query into YES, NO or UNDETERMINED. (a) One real $CO_2$ record from ISODB [1]: points at $p \le P_{\max}/10$ are visible, the rest hidden; six parametric fits (red) against the certified loading band of the Langmuir-site class (blue) and the shape-only band (teal). Chosen at the median farthest-point best-fit error among fully covered test $CO_2$ records with $P_{\max} \le 10$ bar and at least 8 hidden points. (b) Share of hidden points covered on 607 test records versus extrapolation ratio; median band ratios at 5–10× are 3.68 (ours) and 6.85 (shape only). (c) Certified $CO_2/N_2$ selectivity intervals (1 bar, 15:85) for 11 randomly drawn real pairs, model answers (red ticks), and the answer to $S \ge 50$ (? = UNDETERMINED).

***Abstract*—Mixture screening queries pure-gas isotherm databases via Ideal Adsorbed Solution Theory (IAST), fitting one model per record or comparing an ensemble of fitted families. In-window fits diverge at query pressures because IAST integrates spreading pressure from zero and evaluates the weaker gas beyond its window. Even a grid enumeration of dual-site Langmuir parameters within tolerance spans 96% of the certified selectivity width, while a six-family ensemble spans 26%. We formulate screening as certain-answer evaluation over nonparametric hinge and Langmuir-site Stieltjes classes under a tolerance and Henry cap. The exact certified interval reduces to two few-site envelopes per record, while our extrapolation-horizon and log-Henry laws bound certification beyond the window and without a cap. ISOCERT compiles admitted records once with a continuum certificate and answers interval queries from tables in 0.15 ms (8118× faster than direct evaluation). Online, it evaluates a threshold predicate for YES, NO, or UNDETERMINED from four lookups and computes critical Henry caps and next-measurement advice by direct LPs. Across 13,183 ISODB isotherms and 1,930 pairs, certified bands cover 97.0% of held-out points ($\le 10$ bar) and 96.5% at 5–10× extrapolation, where best-fit models stay within tolerance on 6.9%. Under synthetic in-class truth that respects the cap, ISOCERT makes 0 false decisions across 1,592 certified answers.**

***Index Terms*—scientific data management, certain answers, incomplete data, moment problems, adsorption isotherms, query-driven data acquisition**

## I. INTRODUCTION

Screening porous materials for gas separation queries tabulated pure-gas isotherms via Ideal Adsorbed Solution Theory (IAST) [2], [3]. In the NIST/ARPA-E Isotherm Database (ISODB) [1], each record $T = \{(p_i, q_i)\}_{i=1}^{n}$ stores loading $q_i$ at pressures $0 < p_1 < \cdots < p_n =: P_{\max}$ and fixed temperature. Because mixture data are scarce [4], pipelines pair pure-gas records $T_A, T_B$ on one adsorbent and solve IAST for binary selectivity $S$ at pressure $P$ and mole fraction $y$.

Current workflows fit one parametric model per record or compare an uncertified ensemble of well-fitting families, yet in-window fit quality does not determine the query answer [5]–[8]. Tools such as pyIAST [5] and pyGAPS [6] fit Langmuir [9], dual-site Langmuir, Sips [10], Tóth [11], UNILAN [12], and Freundlich models and pass the lowest-RMSE fit to IAST. When points above $P_{\text{cut}} = P_{\max}/10$ are held out on 607 ISODB records (Fig. 1(a)–(b)), in-window fits within tolerance diverge at higher pressures $r$. At extrapolation ratios $\rho = r/P_{\text{cut}} \in (5, 10]$, the lowest-RMSE fit stays within tolerance on 6.9% of hidden points, and the six-family ensemble covers 58.8%. Across 1,930 real pairs (Fig. 1(c)), the ensemble spans a median factor of 1.36 (26% of the certified width) and correlates

with fit RMSE at 0.21 (Spearman). Even a grid enumeration of the dual-site parameter set ($\mathcal{C}_2$) spans a median of 96% of the certified width (Section VI-E), which point fitting hides.

In-window fits fail because IAST integrates pure-component isotherms from zero pressure and evaluates the weaker gas beyond its measured window. To find the adsorbed fraction $x \in (0,1)$ of gas $A$ at $(P,y)$, IAST equates spreading pressures $\Pi_q(r) = \int_0^r (q(t)/t)\,dt$ at $r_A = Py/x$ and $r_B = P(1-y)/(1-x)$:

$$\Pi_{q_A}(Py/x) = \Pi_{q_B}\big(P(1-y)/(1-x)\big), \quad S = \frac{x/(1-x)}{y/(1-y)}.$$

For the stronger gas $A$, $r_A = Py/x \geq Py$ usually lies in-window, but $\Pi_{q_A}(r_A)$ integrates $q_A(t)/t$ over $[0, p_1]$ governed by the Henry slope $q_A'(0^+)$. The weaker gas $B$ is evaluated at $r_B = P(1-y)/(1-x)$, which exceeds $P_{\max}$ in 59% of real pairs (median ratio 1.73).

To replace point estimates with hard guarantees, we formulate mixture screening as a *certain-answer* query problem [13]–[15] over nonparametric *site-mixture classes*. Each record $(T, e, H)$ with tolerance $e_i = \varepsilon \max_j q_j$ and Henry cap $H \geq q'(0^+)$ defines $\mathcal{C}_\phi(T, e, H)$, the isotherms $q(p) = \int_{[0,\infty)} \phi_b(p)\,d\nu(b)$ ($b \geq 0$) with $|q(p_i) - q_i| \leq e_i$ and nonnegative site measure $\nu([0,\infty)) \leq H$. The step kernel $\phi_b(p) = \min(p, 1/b)$ yields the *hinge class* $\mathcal{C}_{\mathrm{h}}$ (shape only, concave nondecreasing isotherms with $q(0) = 0, q'(0^+) \leq H$). The Langmuir kernel $\phi_b(p) = p/(1+bp)$ yields the *Langmuir-site class* $\mathcal{C}_{\mathrm{L}}$ (complete Bernstein/Stieltjes [10], [16]), containing single- and multi-site Langmuir, UNILAN, and Tóth isotherms, plus Sips and Freundlich in its closure. A threshold query $S \geq \tau$ returns YES if $S \geq \tau$ for all $(q_A, q_B) \in \mathcal{C}_A \times \mathcal{C}_B$, NO if $S < \tau$ for all pairs, and UNDETERMINED otherwise, alongside the *certified interval* $[S^-, S^+]$.

Computing exact certain answers over site-mixture classes poses three challenges. First, no single pessimistic isotherm serves all queries, because the horizon $r_A = Py/x$ depends on the equilibrium $x$, $(P,y)$, and partner gas $B$. Second, each class $\mathcal{C}_\phi(T,e,H)$ is an infinite-dimensional space of measures coupled through the nonlinear implicit IAST equation, so per-query joint optimization is prohibitive. Third, when a query returns UNDETERMINED, the system must separate what no finite table can certify from which new measurement narrows $[S^-, S^+]$.

We resolve these challenges via an exact reduction and two laws that govern system design. First, the *envelope reduction theorem* (Theorem 1) shows that extreme equilibria $x^+ = X(U_A, L_B)$ and $x^- = X(L_A, U_B)$ over $\mathcal{C}_A \times \mathcal{C}_B$ depend only on per-record spreading-pressure *envelopes* $U(r), L(r)$, attained by $(n+2)$-site isotherms (Theorem 2). Second, on Henry-like tables, the *extrapolation-horizon law* (Theorem 3) proves that certification beyond $P_{\max}$ is bounded by a *certification horizon* of order $P_{\max}/\varepsilon$ for all $n$. At low pressure, the *log-Henry law* (Theorem 4) proves that under tolerance slack, $U(r; H)$ diverges without a Henry cap and grows logarithmically in $H$ governed by the *hidden capacity* $c_0$.

Our main contributions are:

- **Certain-answer semantics and envelope reduction (Sections III–IV-B).** To our knowledge, we formulate the first certain-answer semantics for IAST screening over site-mixture classes $(\mathcal{C}_{\mathrm{h}}, \mathcal{C}_{\mathrm{L}})$ and reduce $[S^-, S^+]$ to two $(n+2)$-site envelopes per record (Theorems 1–2).
- **Two laws of isotherm certification (Sections IV-C–IV-D).** We prove the extrapolation-horizon law on Henry-like tables (Theorem 3) at scale $P_{\max}/\varepsilon$ for any $n$, and the log-Henry law under tolerance slack (Theorem 4) governed by $c_0$.
- **The ISOCERT compilation and query engine (Section V).** Following LP query bounding [17], ISOCERT admits records via a misfit-floor LP, computes envelopes with a continuum certificate (Lemma 4), and compiles sound tables (Proposition 1). Online, table lookup takes 0.15 ms (8118× faster than direct evaluation), a threshold predicate decides queries, and direct LPs compute critical caps and lookahead advice.
- **Evaluation on NIST ISODB (Section VI).** Across 13,183 isotherms and 1,930 pairs against a computed dual-site parameter set ($\mathcal{C}_2$), certified bands cover 97.0% of held-out points at $P_{\max} \leq 10$ bar and expose a high-pressure boundary above 10 bar. Under cap-respecting truth, ISOCERT makes 0 false decisions across 1,592 certified answers, and we check 2,512 points on measured mixtures from 109 publications.

## II. Related Work

**IAST screening workflows and model sensitivity.** Mixture screening applies IAST [2], [3] with fast solvers [18]–[20] to NIST ISODB [1] and BISON [4], and returns uncertified point predictions alongside studies of experimental variability [21], [22] and non-ideal IAST departures [23]. Packages such as pyIAST [5], pyGAPS [6], and IAST++ [24] fit one model or interpolate linearly; above $P_{\max}$, pyIAST errors unless given a constant fill (in $\mathcal{C}_{\mathrm{h}}$). Current IAST tools also omit the uncertified dual-site parameter set ($\mathcal{C}_2$), which we include as a baseline. Sensitivity studies show qualitatively that model choice or extrapolation alters IAST rankings [8], [25]–[27], and Gharagheizi et al. [7] resample $\pm 2.5\%$ noise around interpolated tables under ad hoc tails. To our knowledge, no prior work bounds IAST selectivity over a nonparametric class.

**Heterogeneous adsorption and Stieltjes representations.** Classical adsorption theory models energetic heterogeneity by integrating a local kernel against a nonnegative affinity measure [28]–[32], which we use for forward bounding. The Langmuir kernel [9] generates multi-site Langmuir, UNILAN [12], and Tóth [11] models, plus Sips and Freundlich in its Stieltjes and complete-Bernstein closure [10], [16], [33]. Inverse methods fit the site distribution via regularization [34], [35] or linear/quadratic programming [36], while heterogeneous-IAST [37], [38] assumes a known site distribution to alter the mixing rule. We retain standard IAST without inverting the site distribution and optimize linear spreading-pressure functionals over feasible site measures.

**Moment problems, Stieltjes bounds, and partial identification.** Our envelopes specialize Markov–Krein moment theory, Tchebycheff systems, and measure extreme points [39]–[41] with semi-infinite conic duality [42]–[44] to implicit adsorption equilibria. Finite Stieltjes and shape constraints likewise yield Padé envelopes [45]–[48], Bergman–Milton bounds [49]–[52], Gini curve bounds [53], [54], and partial identification [55], [56]. While the moment and Stieltjes machinery is classical, its specialization to IAST queries, the envelope reduction, the two laws, and the compiled engine are ours.

**Incomplete databases, model-based views, and LP query bounding.** Database work studies certain answers over incomplete relations [13]–[15], [57], [58], probabilistic databases [59]–[61], and model-based views [62]–[64] that query one fitted model. Deterministic bounds without distributional priors cover missing data [65], uncertainty-annotated execution [66], [67], and certain ML predictions [68], [69]. Related work bounds risk over monotone quantile functions from observed losses [70] and LP join cardinalities [17], [71]–[75]. Following LpBound [17], we derive a continuum certificate via LP duality, but bound an implicit nonlinear equilibrium over a continuous function cone and compile semi-infinite LP envelopes offline.

## III. Problem Statement

### A. Records and the IAST Query

In NIST ISODB [1], a pure-gas *record* $T = \{(p_i, q_i)\}_{i=1}^{n}$ tabulates loadings $q_i \geq 0$ (mmol/g) at pressures $0 < p_1 < \cdots < p_n =: P_{\max}$ (bar) for one adsorbent and temperature. An uncertain record $(T, e, H)$ adds pointwise *tolerances* $e_i = \varepsilon \max_{1\leq j\leq n} q_j$ ($\varepsilon > 0$) for measurement and digitization error [21], [22] and a *Henry cap* $H > 0$ with $q'(0^+) \leq H$. We set $H = \kappa K_{\min}$ ($\kappa \geq 1$) from the smallest Henry constant $K_{\min}$ consistent with $(T, e)$.

A binary query specifies gases $A, B$ for the same adsorbent and temperature, total pressure $P > 0$, and gas mole fraction $y_A = y \in (0, 1)$ ($y_B = 1 - y$). For pure-gas isotherms $q_A, q_B : [0, \infty) \to [0, \infty)$, Ideal Adsorbed Solution Theory (IAST) [2] equates the reduced *spreading pressures*

$$\Pi_q(r) = \int_0^r \frac{q(t)}{t}\, dt \tag{1}$$

at Raoult pressures $r_A = Py/x$ and $r_B = P(1-y)/(1-x)$ [2], [3]. With $\Pi_A = \Pi_{q_A}$ and $\Pi_B = \Pi_{q_B}$, the adsorbed fraction $x \in (0, 1)$ of $A$ and *selectivity* $S = S(q_A, q_B)$ satisfy

$$\Pi_A\left(\frac{Py}{x}\right) = \Pi_B\left(\frac{P(1-y)}{1-x}\right), \quad S = \frac{x/(1-x)}{y/(1-y)}. \tag{2}$$

Fitting one curve per record is fragile in Eq. (2) because IAST evaluates isotherms outside $[p_1, P_{\max}]$. In our running flue-gas query ($A = \mathrm{CO_2}$, $B = \mathrm{N_2}$, $P = 1$ bar, $y = 0.15$, $P_{\max,A} = P_{\max,B} = 1$ bar, threshold $\tau = 50$), $S = 50$ gives $x \approx 0.898$. There $\mathrm{CO_2}$ is evaluated at $r_A = Py/x \approx 0.17$ bar, where $\Pi_A$ integrates across $(0, p_1)$. Meanwhile, $\mathrm{N_2}$ is evaluated at $r_B = P(1-y)/(1-x) \approx 8.3$ bar, over eight times $P_{\max,B}$. Across 1,930 real pairs, the weaker gas is evaluated above $P_{\max}$ in 59% of pairs, with median *extrapolation ratio* $\rho = r/P_{\max}$ of 1.73.

### B. Site-Mixture Classes

A *site kernel* $\phi_b(p)$ gives the loading of an elementary site with unit Henry slope $\phi_b'(0^+) = 1$, affinity $b \geq 0$, and capacity $1/b$ ($\phi_0(p) = p$ at $b = 0$):

$$\begin{aligned} \phi_b^{\mathrm{L}}(p) &= \frac{p}{1+bp} \qquad \text{(Langmuir site)}, \\ \phi_b^{\mathrm{h}}(p) &= \min\Big(p, \frac{1}{b}\Big) \quad \text{(hinge site)}. \end{aligned} \tag{3}$$

Both kernels are concave and nondecreasing on $[0, \infty)$ with $\phi_b(0) = 0$. Physically, $\phi_b^{\mathrm{L}}$ models localized monolayer adsorption [9], and $\phi_b^{\mathrm{h}}$ models step saturation at $1/b$.

**Definition 1** (Site-Mixture Class)**.** Given a record $T = \{(p_i, q_i)\}_{i=1}^{n}$, tolerances $e = (e_1, \ldots, e_n) > 0$, a Henry cap $H > 0$, and a site kernel $\phi_b$, the *site-mixture class* is generated by finite nonnegative Borel measures $\nu$ on $[0, \infty)$:

$$\mathcal{C}_\phi(T, e, H) = \Big\{\, q(p) = \int_{[0,\infty)} \phi_b(p)\, d\nu(b) \;:\; \nu \geq 0, \\ \nu([0,\infty)) \leq H,\ |q(p_i) - q_i| \leq e_i\ \forall i \,\Big\}.$$

With Henry constant $q'(0^+) = \nu([0, \infty))$, we write $\mathcal{C}_{\mathrm{L}} = \mathcal{C}_{\phi^{\mathrm{L}}}(T, e, H)$ for the *Langmuir-site class* and $\mathcal{C}_{\mathrm{h}} = \mathcal{C}_{\phi^{\mathrm{h}}}(T, e, H)$ for the *hinge class* (shape only). The *dual-site parameter set* ($\mathcal{C}_2 \subset \mathcal{C}_{\mathrm{L}}$) restricts $\nu$ to at most two sites as an uncertified baseline. The *misfit floor* $\varepsilon^\star$ is the smallest full-scale tolerance $\varepsilon$ with $\mathcal{C}_\phi(T, e, \infty) \neq \emptyset$, and records meeting our $\varepsilon^\star$ gate are *admitted records*.

**Lemma 1** (Class Characterization and Contained Families)**.** *(i) The hinge cone* $\big\{\int_{[0,\infty)} \phi_b^{\mathrm{h}}(p)\, d\nu(b) : \nu \geq 0,\ \nu([0,\infty)) \leq H\big\}$ *equals the set of concave, nondecreasing* $q : [0, \infty) \to [0, \infty)$ *with* $q(0) = 0$ *and* $q'(0^+) \leq H$*. (ii) A function* $q$ *lies in the Langmuir-kernel cone with finite Henry constant if and only if* $q(p)/p$ *is a Stieltjes function with a finite limit at* $0^+$*, that is,* $q$ *is a complete Bernstein function with* $q(0^+) = 0$ *and* $q'(0^+) < \infty$ *[16]. The cone contains Langmuir and multi-site Langmuir isotherms, UNILAN (a uniform site-energy distribution [12]) and the Tóth isotherm* $q = q_s bp\,(1+(bp)^t)^{-1/t}$ *with* $0 < t \leq 1$*: for* $\arg z = \theta \in (0, \pi)$ *we have* $\arg\big(1 + (bz)^t\big) \in (0, t\theta)$*, so* $q(z)/z$ *maps the upper half-plane into the lower half-plane, is positive on* $(0, \infty)$ *and tends to* $q_s b$ *at* $0$*, which characterises such Stieltjes functions [16]. Sips isotherms with exponent in* $(0, 1)$ *and Freundlich isotherms have an infinite Henry constant and lie in the closure of the cone [10], [33]. Every Langmuir-kernel member is concave and nondecreasing, so* $\mathcal{C}_{\mathrm{L}}(T, e, H) \subseteq \mathcal{C}_{\mathrm{h}}(T, e, H)$.

*Proof.* For (i), if $q$ is concave and nondecreasing with $q(0) = 0$ and $q'(0^+) \leq H$, its right derivative $q'$ is nonnegative and nonincreasing on $(0, \infty)$. Pushing forward $-dq'(s)$ on saturation pressures $s = 1/b \in (0, \infty)$ and placing mass $q'(\infty)$ at $b = 0$ defines $\nu \geq 0$ with $\int_{[0,\infty)} \min(p, 1/b)\, d\nu(b) = \int_0^p \nu([0, 1/t))\, dt = \int_0^p q'(t)\, dt = q(p)$ and $\nu([0, \infty)) = q'(0^+) \leq H$; the converse holds since each $\phi_b^{\mathrm{h}}$ is concave and nondecreasing. For (ii), the equivalence with complete Bernstein functions with $q(0^+) = 0$ and $q'(0^+) = \nu([0, \infty)) <$

$\infty$ is the Stieltjes representation [16], and part (i) gives $\mathcal{C}_{\mathrm{L}} \subseteq \mathcal{C}_{\mathrm{h}}$. □

The hinge class $\mathcal{C}_{\mathrm{h}}$ imposes concavity and monotonicity alone, including pyIAST's piecewise-linear constant-fill interpolant [5] on concave data. By contrast, $\mathcal{C}_{\mathrm{L}}$ models heterogeneous adsorption over nonnegative site-energy distributions [28]–[32]. Across 18 Tóth [11] ($t \in [0.15, 1]$), 15 Sips (exponent $\in [0.2, 1]$), 12 UNILAN, and 3 Freundlich settings (40 exact samples over 4.5 decades), the best site-mixture fit has full-scale misfit at most $2 \times 10^{-4}$. An S-shaped Sips curve with exponent 1.6 (outside the class) has misfit floor 8.3%.

By Tonelli's theorem, $\Pi_q(r) = \int_{[0,\infty)} \psi_r(b)\, d\nu(b)$ is a *linear functional* of $\nu$ with kernel $\psi_r(b) = \int_0^r (\phi_b(t)/t)\, dt$ and $\psi_r(0) = r$. Here $\psi_r(b) = \ln(1+br)/b$ for $\mathcal{C}_{\mathrm{L}}$, and $\psi_r(b) = r$ ($br \le 1$) or $(1+\ln(br))/b$ ($br > 1$) for $\mathcal{C}_{\mathrm{h}}$. For every non-zero $q \in \mathcal{C}_\phi(T, e, H)$, $\Pi_q$ is continuous and strictly increasing on $(0,\infty)$ with $\Pi_q(0^+) = 0$ and $\Pi_q(\infty) = \infty$. Each $\Pi_q(r)$ is convex in $\ln r$ since $d\Pi_q/d(\ln r) = q(r)$ is nondecreasing.

### C. Certain Answers

Under incomplete-database semantics [13]–[15], [57], $\mathcal{C}_A \times \mathcal{C}_B = \mathcal{C}_\phi(T_A, e_A, H_A) \times \mathcal{C}_\phi(T_B, e_B, H_B)$ is the set of possible worlds. At $(P, y)$, the *possible answers* $\{S(q_A, q_B) : (q_A, q_B) \in \mathcal{C}_A \times \mathcal{C}_B\}$ form the *certified interval* $[S^-, S^+]$ with $S^- = \inf_{\mathcal{C}_A \times \mathcal{C}_B} S$ and $S^+ = \sup_{\mathcal{C}_A \times \mathcal{C}_B} S$. A query $S \ge \tau$ has *certain answer* YES if $S \ge \tau$ on $\mathcal{C}_A \times \mathcal{C}_B$, NO if $S < \tau$ on $\mathcal{C}_A \times \mathcal{C}_B$, and UNDETERMINED otherwise (YES at $\tau = 50$ requires $S^- \ge 50$).

Certain answers are certified relative to (i) the binary IAST mixing rule in Eq. (2) [23] and (ii) the stated class $\mathcal{C}_\phi$, tolerance $e$, and Henry cap $H$. For records thinned from $n > 40$, the class and sharpness refer to the retained points. We study three problems: (1) computing $[S^-, S^+]$ from two univariate envelopes per gas (Sections IV-A–IV-B), (2) bounding certification beyond $P_{\max}$ or without a cap (Sections IV-C–IV-D), and (3) compiling admitted records for sub-millisecond screening (Section V).

## IV. Envelopes, Sharp Answers and Two Laws

### A. Envelopes

As our standing assumption, $\mathcal{C} = \mathcal{C}_\phi(T, e, H)$ is nonempty, $H < \infty$, and $q_i - e_i > 0$ for at least one node $i \in \{1, \dots, n\}$ (excluding the zero isotherm). Each record's uncertainty over $\mathcal{C}$ at horizon $r > 0$ is summarized by its upper and lower spreading-pressure *envelopes* $U(r), L(r)$ and loading band $[q_{\mathrm{lo}}(r), q_{\mathrm{hi}}(r)]$:

$$\begin{aligned} U(r) &= \sup_{q \in \mathcal{C}} \Pi_q(r), & L(r) &= \inf_{q \in \mathcal{C}} \Pi_q(r), \\ q_{\mathrm{hi}}(r) &= \sup_{q \in \mathcal{C}} q(r), & q_{\mathrm{lo}}(r) &= \inf_{q \in \mathcal{C}} q(r). \end{aligned} \tag{4}$$

Subscripts $A, B$ denote the mixture gases ($U_A, L_A$ and $U_B, L_B$).

**Lemma 2** (Attainment and Regularity of Envelopes)**.** *For each horizon $r > 0$, the supremum and infimum defining $U(r)$, $L(r)$, $q_{\mathrm{hi}}(r)$, and $q_{\mathrm{lo}}(r)$ in Eq.* (4) *are attained by members of $\mathcal{C}$. Moreover, $U$ and $L$ are $H$-Lipschitz and strictly increasing on $(0,\infty)$, satisfy $\lim_{r\to 0^+} U(r) = \lim_{r\to 0^+} L(r) = 0$ and $\lim_{r\to\infty} U(r) = \lim_{r\to\infty} L(r) = \infty$, and $U(r)$ is convex as a function of $\ln r$. Moreover $q_{\mathrm{lo}}(r) \ge (q_i - e_i)\min(1, r/p_i) > 0$ for every node $i$ with $q_i > e_i$.*

*Proof.* On the one-point compactification $[0,\infty]$, for fixed $p, r > 0$ the maps $b \mapsto \phi_b(p)$ and $b \mapsto \psi_r(b)$ extend continuously with $\phi_\infty(p) = \psi_r(\infty) = 0$. By the Banach–Alaoglu theorem, nonnegative Borel measures on $[0,\infty]$ of mass at most $H$ are weak-* compact, the tolerance constraints $|\int_{[0,\infty]} \phi_b(p_i)\, d\nu(b) - q_i| \le e_i$ are weak-* closed, and the objectives $\nu \mapsto \int_{[0,\infty]} \psi_r(b)\, d\nu(b)$ and $\nu \mapsto \int_{[0,\infty]} \phi_b(r)\, d\nu(b)$ are weak-* continuous, so every extremum in Eq. (4) is attained on $[0,\infty]$. Deleting mass at $b = \infty$ changes neither $q(p)$ nor $\Pi_q(r)$ and does not increase total mass, yielding an optimal $q^\circ \in \mathcal{C}$.

Because $|d\psi_r(b)/dr| = \phi_b(r)/r \le 1$, for every $q \in \mathcal{C}$ the spreading pressure $\Pi_q$ is $H$-Lipschitz in $r$, which passes to $U$ and $L$. A member is nondecreasing, and concave with $q(0) = 0$, so $q(r) \ge q(p_i)\min(1, r/p_i)$; taking the infimum gives $q_{\mathrm{lo}}(r) \ge (q_i - e_i)\min(1, r/p_i) > 0$ whenever $q_i > e_i$. For $0 < r < r'$, if $q^\circ \in \mathcal{C}$ attains $U(r)$, then $U(r') \ge \Pi_{q^\circ}(r') > \Pi_{q^\circ}(r) = U(r)$; if $q_\circ \in \mathcal{C}$ attains $L(r')$, then $L(r') = \Pi_{q_\circ}(r') > \Pi_{q_\circ}(r) \ge L(r)$. Since $0 \le L(r) \le U(r) \le Hr$, both vanish as $r \to 0^+$. For $i$ with $q_i > e_i$ and $r > p_i$, $\Pi_q(r) \ge \int_{p_i}^r (q(t)/t)\, dt \ge (q_i - e_i)\ln(r/p_i) \to \infty$ as $r \to \infty$, so $L(r), U(r) \to \infty$. Finally, since each $\Pi_q(r)$ is convex in $\ln r$, their supremum $U(r)$ is convex in $\ln r$. □

### B. Envelope Reduction

At $P > 0$ and $y \in (0,1)$, bivariate optimization over $\mathcal{C}_A \times \mathcal{C}_B$ decouples into univariate envelopes because the IAST root $X(\Pi_A, \Pi_B)$ is monotone in $(\Pi_A, \Pi_B)$.

**Lemma 3** (IAST Well-Posedness and Monotonicity)**.** *(i) For any continuous, strictly increasing $\Pi_A, \Pi_B : (0,\infty) \to (0,\infty)$ with $\Pi_i(0^+) = 0$ and $\Pi_i(\infty) = \infty$, the equation $\Pi_A(Py/x) = \Pi_B(P(1-y)/(1-x))$ has a unique solution $x = X(\Pi_A, \Pi_B) \in (0,1)$. (ii) If $\Pi_A \le \tilde{\Pi}_A$ and $\tilde{\Pi}_B \le \Pi_B$ pointwise on $(0,\infty)$, then $X(\Pi_A, \Pi_B) \le X(\tilde{\Pi}_A, \tilde{\Pi}_B)$.*

*Proof.* For (i), as $x$ runs over $(0,1)$, $\Pi_A(Py/x)$ decreases continuously and strictly from $\infty$ to $\Pi_A(Py) \in (0,\infty)$, while $\Pi_B(P(1-y)/(1-x))$ increases continuously and strictly from $\Pi_B(P(1-y)) \in (0,\infty)$ to $\infty$; their difference has a unique root $X(\Pi_A, \Pi_B) \in (0,1)$. For (ii), at $x_0 = X(\Pi_A, \Pi_B)$ we have $\tilde{\Pi}_A(Py/x_0) \ge \Pi_A(Py/x_0) = \Pi_B(P(1-y)/(1-x_0)) \ge \tilde{\Pi}_B(P(1-y)/(1-x_0))$. Since $x \mapsto \tilde{\Pi}_A(Py/x) - \tilde{\Pi}_B(P(1-y)/(1-x))$ is strictly decreasing and nonnegative at $x_0$, $X(\tilde{\Pi}_A, \tilde{\Pi}_B) \ge x_0$. □

**Theorem 1** (Envelope Reduction; the Certified Interval Is Sharp)**.** *Let $\mathcal{C}_A$ and $\mathcal{C}_B$ satisfy the standing assumption, with*

*spreading-pressure envelopes* $(U_A, L_A)$ *and* $(U_B, L_B)$*. For any* $P > 0$ *and* $y \in (0,1)$*,*

$$\sup_{(q_A,q_B)\in\mathcal{C}_A\times\mathcal{C}_B} x = X(U_A, L_B) =: x^+,$$
$$\inf_{(q_A,q_B)\in\mathcal{C}_A\times\mathcal{C}_B} x = X(L_A, U_B) =: x^-.$$

*Both extrema are attained by pairs in* $\mathcal{C}_A \times \mathcal{C}_B$*, and the set of possible selectivities* $\{S(q_A, q_B) : q_A \in \mathcal{C}_A,\ q_B \in \mathcal{C}_B\}$ *is the exact closed interval* $[S^-, S^+]$ *with*

$$S^\pm = \frac{x^\pm/(1-x^\pm)}{y/(1-y)}.$$

*Proof.* By Lemma 2, $U_A, L_A, U_B, L_B$ satisfy Lemma 3(i), so $x^+ = X(U_A, L_B)$ and $x^- = X(L_A, U_B)$ lie in $(0,1)$, and Lemma 3(ii) gives $x^- \le X(\Pi_{q_A}, \Pi_{q_B}) \le x^+$ for all $(q_A, q_B) \in \mathcal{C}_A \times \mathcal{C}_B$. For attainment of $x^+$, set $r_A = Py/x^+$ and $r_B = P(1-y)/(1-x^+)$, and pick $q_A^\circ \in \mathcal{C}_A$ attaining $U_A(r_A)$ and $q_B^\circ \in \mathcal{C}_B$ attaining $L_B(r_B)$ by Lemma 2. Then $\Pi_{q_A^\circ}(r_A) = U_A(r_A) = L_B(r_B) = \Pi_{q_B^\circ}(r_B)$, so $X(\Pi_{q_A^\circ}, \Pi_{q_B^\circ}) = x^+$ by uniqueness; $x^-$ is symmetric. Along the segment between two pairs of members the spreading pressures are affine in the segment parameter, so the unique root varies continuously and takes every intermediate value. □

*Remark* 1 (No Worst-Case Isotherm, but a Worst-Case Envelope). In Theorem 1, the maximizer $q_A^\circ \in \mathcal{C}_A$ depends on $r_A = Py/x^+$, and thus on $(P, y)$ and partner gas $B$ via $L_B$; no single pessimistic isotherm serves all queries. Instead, the two envelope *functions* $U_A$ and $L_A$ of each record suffice and can be compiled once for all partners and conditions.

**Theorem 2** (Extremal Isotherms Have Few Sites)**.** *For each horizon* $r > 0$*, the values* $U(r)$*,* $L(r)$*,* $q_{\mathrm{hi}}(r)$*, and* $q_{\mathrm{lo}}(r)$ *over* $\mathcal{C}_\phi(T, e, H)$ *are attained by discrete measures* $\nu$ *with at most* $n+2$ *atoms in* $[0,\infty)$*: by ordinary* $(n+2)$*-site isotherms of the class (in* $\mathcal{C}_{\mathrm{L}}$*, a multi-site Langmuir model, possibly with one Henry site at* $b = 0$*).*

*Proof.* Let $\nu^\circ$ be an optimal measure on $[0,\infty)$ from Lemma 2 with mass $M = \nu^\circ([0,\infty)) > 0$ (if $M = 0$, 0 atoms suffice), and define $g : [0,\infty] \to \mathbb{R}^{n+1}$ by $g(b) = (\phi_b(p_1), \dots, \phi_b(p_n), \psi_r(b))$ with $g(\infty) = 0$. Then $M^{-1}\int_{[0,\infty)} g(b)\, d\nu^\circ(b)$ lies in the convex hull of the compact curve $g([0,\infty]) \subset \mathbb{R}^{n+1}$. By Carathéodory's theorem [39]–[41], it equals a convex combination $\sum_{k=1}^m \alpha_k g(b_k)$ of $m \le n+2$ points $b_k \in [0,\infty]$. Dropping any atom at $b_k = \infty$, $\hat\nu = \sum_{b_k<\infty} M\alpha_k \delta_{b_k}$ has at most $n+2$ sites, mass at most $M \le H$, and identical $q(p_i)$ and $\Pi_q(r)$. Replacing $\psi_r(b)$ by $\phi_b(r)$ gives the claim for $q_{\mathrm{hi}}(r)$ and $q_{\mathrm{lo}}(r)$. □

In our runs, extremal measures use 2 to 5 sites. The uncertified dual-site parameter set ($\mathcal{C}_2$) realizes a median of 96% of the certified width on real pairs (Section VI-E).

Because $\mathcal{C}_\phi(T, e, H)$ grows with $H$, $U(r; H)$ is nondecreasing and $L(r; H)$ is nonincreasing in $H$. By Lemma 3(ii) and Theorem 1, $S^+$ is nondecreasing and $S^-$ is nonincreasing in $(H_A, H_B)$. A YES or NO decision certified at cap $H$ holds for all $H' \le H$. Its *critical cap* is the supremum of caps $H$ (or multipliers $\kappa = H/K_{\min}$) where certification holds.

### C. Law 1: Extrapolation Horizon

IAST typically evaluates the weaker gas at $r \gg P_{\max}$, where records on $[p_1, P_{\max}]$ are nearly linear (Henry-like). On a table fitting $q_i = Kp_i$ within full-scale tolerance $\varepsilon$, an adversarial single-site Langmuir curve matches $Kp$ within $\pm\varepsilon K P_{\max}$ on $[0, P_{\max}]$ and saturates above $P_{\max}$.

**Theorem 3** (Extrapolation-Horizon Law)**.** *Consider the Langmuir-site class* $\mathcal{C}_{\mathrm{L}}(T, e, H)$ *for a Henry-like table* $q_i = Kp_i$ *(*$0 < K \le H$*) with full-scale tolerance* $e_i = \varepsilon K P_{\max}$ *(*$0 < \varepsilon < 1/2$*), any* $n \ge 1$*, and any node placement* $0 < p_1 < \cdots < p_n = P_{\max}$*. Define*

$$a^\star(\varepsilon) = u_\star^2 - 1, \qquad u_\star = \frac{1 - \varepsilon + \varepsilon\sqrt{2(1-\varepsilon)}}{1-2\varepsilon},$$

*the positive root of* $(\sqrt{1+a}-1)^2(1-\varepsilon) = \varepsilon(a+\varepsilon)$*, with* $a^\star = (2+2\sqrt{2})\varepsilon + O(\varepsilon^2)$*. Take any* $0 \le a \le a^\star(\varepsilon)$ *with* $(1+a)K \le H$*, and set*

$$b_a = \frac{a+\varepsilon}{(1-\varepsilon)P_{\max}}.$$

*Then the single-site isotherm* $q_a(p) = \frac{(1+a)Kp}{1+b_a p}$ *belongs to* $\mathcal{C}_{\mathrm{L}}(T, e, H)$*, so for every horizon* $r > 0$*,*

$$L(r) \le \frac{(1+a)K}{b_a}\ln(1+b_a r), \qquad U(r) \ge Kr.$$

*Hence*

$$\frac{U(r)}{L(r)} \ge \frac{b_a r}{(1+a)\ln(1+b_a r)},$$

*which is unbounded in* $r$ *and depends on* $T$ *only through* $\varepsilon$ *and* $P_{\max}$*: no number of additional points inside* $(0, P_{\max}]$ *changes it.*

*Proof.* The deviation $d(p) = q_a(p) - Kp = Kp\frac{a - b_a p}{1+b_a p}$ is concave on $[0,\infty)$ (since $d''(p) = -2Kb_a(1+a)(1+b_a p)^{-3} < 0$) with $d(0) = 0$. Its global maximum over $p \ge 0$ is $K(\sqrt{1+a}-1)^2/b_a$, attained where $1 + b_a p = \sqrt{1+a}$; this maximum is at most $\varepsilon K P_{\max}$ if and only if

$$\left(\sqrt{1+a}-1\right)^2 \le \varepsilon b_a P_{\max} = \frac{\varepsilon(a+\varepsilon)}{1-\varepsilon},$$

i.e., if and only if $a \le a^\star(\varepsilon)$. By concavity, the minimum of $d$ on $[0, P_{\max}]$ is at an endpoint: $d(0) = 0$ and, since $1 + b_a P_{\max} = (1+a)/(1-\varepsilon)$, we have $d(P_{\max}) = -\varepsilon K P_{\max}$. Thus $|q_a(p) - Kp| \le \varepsilon K P_{\max}$ on $[0, P_{\max}]$, in particular at every node $p_i$, and the Henry constant is $(1+a)K \le H$. Hence $q_a \in \mathcal{C}_{\mathrm{L}}(T, e, H)$ and $L(r) \le \Pi_{q_a}(r) = (1+a)K\ln(1 + b_a r)/b_a$. Meanwhile, the Henry isotherm $Kp$ belongs to $\mathcal{C}_{\mathrm{L}}(T, e, H)$, giving $U(r) \ge Kr$. □

The witness $q_a$ saturates at capacity $(1+a)K/b_a = KP_{\max}/(c\varepsilon)$ with $c = (a+\varepsilon)/((1+a)(1-\varepsilon)\varepsilon)$, about 1 for known Henry slope ($a = 0$) and $3 + 2\sqrt{2}$ when free ($a = a^\star$, small $\varepsilon$). Past this *certification horizon* $P_{\max}/(c\varepsilon)$ ($\rho = r/P_{\max} \sim 1/(c\varepsilon)$), $L(r)$ is at most logarithmic in $r$

whereas $U(r)$ is at least linear. Theorem 3 upper-bounds $L$ and does not state that $L$ equals the witness. Across 300 numerical configurations (at least six equally spaced nodes, $\rho \geq 3$), the LP-to-witness ratio lies in $[0.9845, 1]$ (median 1.0000; Section VI-F). With few or poorly placed nodes, $L$ can be much smaller. Points in $(0, P_{\max}]$ cannot remove this bound. Only raising $P_{\max}$ or tightening $\varepsilon$ extends the horizon. In $\mathcal{C}_{\mathrm{h}}$, the witness $\min(Kp, KP_{\max})$ yields $L(r) \leq KP_{\max}(1+\ln(r/P_{\max}))$ for $r \geq P_{\max}$ at any tolerance, so the shape-only lower envelope turns logarithmic at $\rho = 1$ rather than $1/(c\varepsilon)$.

### D. Law 2: Log-Henry

Whereas Theorem 3 governs the high-pressure tail via $L(r)$, our second law governs the origin $p \to 0^+$ via $U(r)$. Because measurements start at $p_1 > 0$ with tolerances $e_i > 0$, a small-capacity site can saturate below $p_1$ and inflate $\int_0^r (q(t)/t)\, dt$ near $t = 0$.

**Theorem 4** (Log-Henry law)**.** *Let $q_0$ be an element of the kernel cone with Henry constant $H_0$, and let $c > 0$ satisfy $q_i - e_i < q_0(p_i) + c \leq q_i + e_i$ for all $i$. Put $\beta_0 = \max_i \big(q_i - e_i - q_0(p_i)\big)^+ \big/ \big(p_i\,(q_0(p_i) + c - q_i + e_i)\big)$ for the Langmuir kernel and $\beta_0 = 1/p_1$ for the hinge kernel. Then for every $H \geq H_0 + c\beta_0$ and every $r > 0$, the first bound below holds for Langmuir sites and the second for hinge sites when $(H - H_0)r \geq c$:*

$$U(r; H) \geq \Pi_{q_0}(r) + c\,\ln\Big(1 + \tfrac{(H-H_0)r}{c}\Big),$$
$$U(r; H) \geq \Pi_{q_0}(r) + c\,\Big(1 + \ln\tfrac{(H-H_0)r}{c}\Big),$$

*Conversely, with $u_1 = q_1 + e_1$ and $\varrho = \min(r, p_1)$, whenever $H\varrho \geq u_1$ in either class,*

$$U(r; H) \leq u_1\Big(1 + \ln\frac{H\varrho}{u_1}\Big) + u_1\,\frac{(r - p_1)^+}{p_1}.$$

*Proof.* Add to $q_0$ one site of capacity $c$ and affinity $\beta = (H - H_0)/c \geq \beta_0$. Its response, $c\beta p/(1+\beta p)$ or $c\min(\beta p, 1)$, lies in $[0, c]$, so the upper tolerances hold. For the hinge kernel the site equals $c$ at every node because $\beta p_1 \geq 1$; for the Langmuir kernel $q_0(p_i) + c\beta p_i/(1+\beta p_i) \geq q_i - e_i$ holds exactly when $\beta \geq (q_i - e_i - q_0(p_i))^+/(p_i(q_0(p_i) + c - q_i + e_i))$. The Henry constant is $H_0 + c\beta = H$, so the new curve is a member, and its spreading pressure exceeds $\Pi_{q_0}(r)$ by $c\ln(1 + \beta r)$, respectively $c(1 + \ln(\beta r))$ when $\beta r \geq 1$. For the upper bound, every member has slope at most $H$ and is nondecreasing with $q(p_1) \leq u_1$, so $q(t) \leq \min(Ht, u_1)$ on $(0, p_1]$; and $q(t)/t$ is nonincreasing, so $q(t) \leq u_1 t/p_1$ for $t \geq p_1$. Integrating $q(t)/t$ gives the claim. □

Whenever such $(q_0, c)$ exists, $U(r; H)$ grows logarithmically in $H$ (between $c\ln H$ and $u_1 \ln H$ up to additive terms) and diverges as $H \to \infty$ for every $r$. By Theorem 1, $S^+ \to \infty$ as $H_A \to \infty$ and $S^- \to 0$ as $H_B \to \infty$, so without a Henry cap such a table certifies no upper bound. Tolerance slack is necessary because for $p = (1, 2)$, $q = (0.9, 2.1)$, $e = (0.1, 0.1)$, concavity forces $q(p) = p$ on $[0, 2]$ and $U(r; H) = r$ for all $H \geq 1$. The increment does not depend on $p_1$, and a new low-pressure point at the same tolerance removes at most part of $c$. The *hidden capacity* $c_0 = \max\{c : \exists\, q_0$ in the cone, $|q_0(p_i) + c - q_i| \leq e_i\ \forall i\}$ (one LP) bounds admissible $c$. Theorem 4 proves logarithmic bounds, whereas matching the asymptotic slope to $c_0$ is empirical. Across 857 real records, $(\mathrm{d}U/\mathrm{d}\ln H)/c_0$ has median 0.9920 (quartiles 0.924–0.9999; Section VI-F, Fig. 4(b)).

## V. The IsoCert System

### A. Overview and Admission

IsoCert turns Theorems 1–4 into a two-phase engine that admits records and compiles envelopes offline (Stages 1–3) to answer mixture queries online (Stages 4–5; Fig. 2). Because Theorems 1–2 reduce binary queries to two $(n + 2)$-site envelopes $U(r)$ and $L(r)$ per record, our running $CO_2/N_2$ records ($P = 1$ bar, $y = 0.15$, $\tau = 50$) are compiled once and reused. Before compilation, IsoCert thins $n > 40$ records to 40 evenly spaced points by index and solves one LP on a dense affinity grid for the *misfit floor* $\varepsilon^\star$ (Section III-B). Records with $\varepsilon^\star \leq 3\%$ are *admitted records* (68.5% of 13,183 parsed ISODB records under $\mathcal{C}_{\mathrm{L}}$ and 76.6% under $\mathcal{C}_{\mathrm{h}}$); the rest are rejected.

For each admitted record, IsoCert sets a working tolerance and Henry cap, with

$$\varepsilon = \max(m\,\varepsilon^\star,\, 1\%), \qquad e_i = \varepsilon \max_{1 \leq j \leq n} q_j, \tag{5}$$

where the $1\%$ floor guards smooth records and $m = 3$ is calibrated on a $30\%$ split (Section VI-A). Because $U(r; H)$ diverges as $H \to \infty$ under tolerance slack (Theorem 4), IsoCert solves one LP at tolerance $e_i$ for $K_{\min} = \min\{\nu([0, \infty)) : \nu \geq 0,\ |\int \phi_b(p_i)\, d\nu(b) - q_i| \leq e_i\ \forall i\}$ and sets $H = \kappa K_{\min}$ with default $\kappa = 10$. Computing $\varepsilon^\star$ and $K_{\min}$ on the affinity grid yields upper bounds on the continuous optima, which only fix the class inputs $e_i$ and $H$ of the guarantee.

### B. Certified Envelope LP

Following semi-infinite duality [42]–[44] and LP database bounds [17], Algorithm 1 solves a grid LP in $u_b \geq 0$ (site-$b$ loading at $P_{\max}$) with column generation and returns a dual bound. Although Theorem 2 guarantees attainment by at most $n+2$ sites, optimizing affinities directly is non-convex. Solving only over a finite grid $B$ (initialized at 8 affinities per decade on $[b_{\min}, b_{\max}] = [10^{-5}/P_{\max}, 10^3\kappa/p_1]$ plus $b = 0$) is an inner approximation that can under-estimate $U(r)$ and over-estimate $L(r)$. Rescaling by $w_b = \phi_b(P_{\max})$ gives $q(p_i) = \sum_b a_i(b) u_b$ with data coefficients $a_i(b) \in (0, 1]$ for all $H$, Henry constraint $\sum_b u_b/(Hw_b) \leq 1$, and objective $\sum_b c(b) u_b$ (with $\phi_b(r)$ replacing $\psi_r(b)$ for $q_{\mathrm{hi}}, q_{\mathrm{lo}}$). For $r > r_c$ (Line 1), the far-field continuations hold for every member because $q$ is nondecreasing and $q(t)/t$ is nonincreasing.

**Lemma 4** (Continuum certificate)**.** *Let $\lambda \in \mathbb{R}^{2n+1}_{\geq 0}$ be arbitrary, $D = -d^\top \lambda$, and $\mathrm{rc}(b) = c(b) + A(b)^\top \lambda$ for $b \in [0, \infty)$. If $v \geq 0$ satisfies $\mathrm{rc}(b) \geq -v$ for every $b \geq 0$, then $s\,\Pi_q(r) \geq D - vM$ for every $q \in \mathcal{C}_\phi(T, e, H)$; that is, $L(r) \geq D - vM$ for $s = +1$ and $U(r) \leq -(D - vM)$ for $s = -1$.*

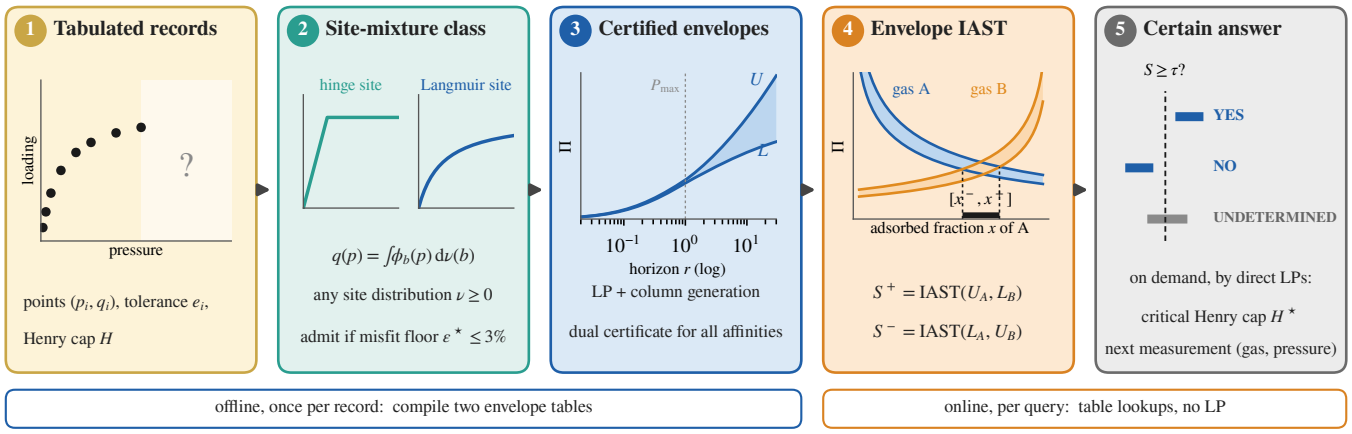



Fig. 2. IsoCert compiles pure-gas records offline into certified envelopes and answers mixture queries online without linear programming. Offline (Stages 1–3): (1) record tolerance and Henry cap, (2) site-mixture class and misfit-floor admission, (3) envelope LP with column generation and a continuum certificate. Online (Stages 4–5): (4) envelope IAST for $[S^-, S^+]$, (5) YES/NO/UNDETERMINED from table lookups; the critical cap and measurement advice are computed on demand by direct LPs. Mini-plots are schematic (blue: Langmuir-site; teal: hinge; red: fits; black: data).

**Algorithm 1** Envelope LP with Column Generation and Continuum Certificate

**Require:** Record $T$, tolerances $e_i$, Henry cap $H$, kernel $\phi_b$, horizon $r > 0$, sense $s \in \{+1, -1\}$ ($+1$: $L(r)$, $-1$: $U(r)$), affinity grid $B$

**Ensure:** $\widetilde{V}$ with $\widetilde{V} \leq L(r)$ if $s = +1$, $\widetilde{V} \geq U(r)$ if $s = -1$, for the continuous class

**if** $r > r_c := 10^4 P_{\max}$ **then return** $L(r_c) + q_{\mathrm{lo}}(r_c)\ln(r/r_c)$ ($s = +1$) or $U(r_c) + q_{\mathrm{hi}}(r_c)(r - r_c)/r_c$ ($s = -1$)
$w_b \leftarrow \phi_b(P_{\max})$; $c(b) \leftarrow s\,\psi_r(b)/w_b$
$a_i(b) \leftarrow \phi_b(p_i)/w_b$; $A(b) \leftarrow \big((a_i)_{i\leq n}, (-a_i)_{i\leq n}, \frac{1}{Hw_b}\big)^\top$
$d \leftarrow \big((q_i + e_i)_{i\leq n}, -(q_i - e_i)_{i\leq n}, 1\big)^\top$; $M \leftarrow q_n + e_n$
**for** round $= 1, \ldots, 30$ **do**
  Solve $\min_{u\geq 0}\{\sum_{b\in B} c(b)u_b : \sum_{b\in B} A(b)u_b \leq d\}$; $\lambda \leftarrow$ multipliers clipped at 0; $D \leftarrow -d^\top\lambda$
  $(v, \delta, B^+) \leftarrow$ CERTIFY$(\lambda, \theta)$ with $\theta = \frac{1}{2}10^{-6}|D|/M$ ▷ $\mathrm{rc}(b) + \delta/(Hw_b) \geq -v$ for all $b \geq 0$
  **if** $vM + \delta \leq 10^{-6}|D|$ **or** $B^+ = \emptyset$ **or** round $= 30$ **then break**
  $B \leftarrow B \cup B^+$ ▷ affinities with $\mathrm{rc} < -\theta$
**end for**
**return** $s\,(D - \delta - vM)$

*Proof.* Let $\nu$ represent $q$ and put $du(b) = \phi_b(P_{\max})\, d\nu(b)$, a nonnegative measure with $\int du = q(P_{\max}) \leq M$. Membership of $q$ reads $\int A(b)\, du(b) \leq d$ componentwise, and $s\,\Pi_q(r) = \int c(b)\, du(b)$. Since $\lambda \geq 0$,

$$\int c\, du = \int \mathrm{rc}\, du - \lambda^\top \int A\, du \;\geq\; -vM - \lambda^\top d \;=\; D - vM.$$

□

Lemma 4 needs no optimality of $\lambda$ and no grid because any nonnegative multipliers and valid $v$ give a *continuum certificate* for every member. The LP supplies $\lambda$; CERTIFY supplies $v$. For the Langmuir kernel, with $\rho_i = p_i/P_{\max}$ and $\ell(x) = \ln(1 + x)/x$,

$$a_i(b) = 1 - \frac{1 - \rho_i}{1 + bp_i}, \qquad \frac{1}{Hw_b} = \frac{1 + bP_{\max}}{HP_{\max}}, \qquad \frac{\psi_r(b)}{w_b} = \frac{r}{P_{\max}}\,\ell(br) + \ln(1 + br). \tag{6}$$

Here $a_i$ and the mass coefficient $1/(Hw_b)$ are nondecreasing in $b$, $\ell(br)$ is decreasing, and $\ln(1+br)$ is increasing. On any interval $[\underline{b}, \overline{b}]$, each term of the reduced cost rc is therefore bounded below at one endpoint chosen by its multiplier's sign. A midpoint mean-value bound $\mathrm{rc}(b_c) - \frac{1}{2}\ln(\overline{b}/\underline{b})\,\sup|\mathrm{d\,rc}/\mathrm{d}\ln b|$ at $b_c$, with the derivative enclosed similarly, is second-order accurate. The hinge kernel is treated piecewise between its kinks $1/p_i$ and $1/r$.

CERTIFY starts from 128 intervals per decade over the column range and bisects intervals with bound below $-\theta$. Bisection stops when all intervals pass or a point with $\mathrm{rc} < -\theta$ yields a new column. The interval $[0, b_{\min}]$ is covered by the one-end bound. Beyond the column range, interval bounds cover twelve further decades and an analytic tail covers $[B, \infty)$. On $[B, \infty)$, $a_i \leq 1$ and only $-\ln(1+br)$ in upper envelopes is unbounded below. The mass multiplier $\lambda_m$ dominates that term because $\lambda_m b/H - \ln(1 + br)$ is convex in $b$ with minimizer $\max(B, H/\lambda_m - 1/r)$. If the tail fails (for example $\lambda_m = 0$ at far horizons), raising $\lambda_m$ by the smallest $\delta$ that makes it pass remains admissible in Lemma 4 and lowers $D$ by $\delta$. On 300 random real pairs, the *certificate margin* $(vM + \delta)/|D|$ is at most $8.7 \times 10^{-7}$ for $\mathcal{C}_{\mathrm{L}}$ and $7.7 \times 10^{-7}$ for $\mathcal{C}_{\mathrm{h}}$ at the horizons determining $[S^-, S^+]$ (100.0% of pairs below $10^{-6}$). Across all 75,767 bisection LPs, 22 stop above $10^{-6}$ (largest $1.9 \times 10^{-6}$), and the margin is subtracted in every case. Bisection depth is at most 19. Relative to a check-grid-only certificate, interval endpoints move by at most $9.2 \times 10^{-4}$ in $\ln S$ and solve time grows by a factor 2.1. Solves use a median of 366 columns per pair (188 from column generation) and 2 to 5 active sites.

Double-precision arithmetic adds a heuristic outward relative slack of $2 \times 10^{-14}$ to every bound. Because we do not use verified interval arithmetic, the exact-arithmetic certificate of Lemma 4 and the floating-point implementation are distinct. Decisions within that slack could be post-checked with verified arithmetic, which we have not done.

### C. Compiled Tables and Query Evaluation

Because $U(r)$ and $L(r)$ depend on a single record, IsoCert precomputes each record on a logarithmic horizon grid and interpolates conservatively so online queries need no LPs (Proposition 1).

**Proposition 1** (Sound compilation)**.** *Let $0 < r_1 < \cdots < r_m$ be a grid of horizons. (i) The piecewise-linear interpolant $\widetilde{U}(r)$ of $\big(\ln r_j,\, U(r_j)\big)$ in $\ln r$ satisfies $\widetilde{U}(r) \geq U(r)$ for all $r \in [r_1, r_m]$. (ii) The piecewise-logarithmic function*

$$\widetilde{L}(r) = L(r_j) + q_{\mathrm{lo}}(r_j) \ln(r/r_j), \qquad r \in [r_j, r_{j+1}), \quad (7)$$

*satisfies $\widetilde{L}(r) \leq L(r)$ for all $r \geq r_1$. Both parts remain valid when the node values are replaced by the certified bounds of Algorithm 1 (upper bounds of $U(r_j)$, lower bounds of $L(r_j)$ and $q_{\mathrm{lo}}(r_j)$). The interpolant $\widetilde{L}$ may jump upward at grid points. For monotone, possibly discontinuous arguments we define $X(\Pi_A, \Pi_B) = \sup\{x : \Pi_A(Py/x) \geq \Pi_B(P(1-y)/(1-x))\}$, which is what bisection returns; Lemma 3(ii) holds verbatim for it. Outside the grid the engine evaluates the envelopes by the LP directly. Consequently, solving Eq.* (2) *with $\widetilde{U}$ and $\widetilde{L}$ in place of $U$ and $L$ yields $[\widetilde{S}^-, \widetilde{S}^+] \supseteq [S^-, S^+]$.*

*Proof.* For (i), $U(r)$ is convex in $\ln r$ (Lemma 2), so its secant $\widetilde{U}(r)$ on $[\ln r_j, \ln r_{j+1}]$ satisfies $\widetilde{U}(r) \geq U(r)$. For (ii), any $q \in \mathcal{C}_\phi(T, e, H)$ is nondecreasing on $[r_j, r]$ with $\Pi_q(r_j) \geq L(r_j)$ and $q(r_j) \geq q_{\mathrm{lo}}(r_j)$, so

$$\begin{aligned}\Pi_q(r) = \Pi_q(r_j) + \int_{r_j}^{r} \frac{q(t)}{t}\, dt &\geq L(r_j) + q(r_j) \ln \frac{r}{r_j} \\ &\geq L(r_j) + q_{\mathrm{lo}}(r_j) \ln \frac{r}{r_j}.\end{aligned}$$

Taking the infimum over $q \in \mathcal{C}_\phi(T, e, H)$ gives $L(r) \geq \widetilde{L}(r)$. Since $\widetilde{U}_A \geq U_A$, $\widetilde{L}_B \leq L_B$ and $\widetilde{L}_A \leq L_A$, $\widetilde{U}_B \geq U_B$ pointwise, Lemma 3(ii) yields $X(\widetilde{L}_A, \widetilde{U}_B) \leq X(L_A, U_B) = x^-$ and $x^+ = X(U_A, L_B) \leq X(\widetilde{U}_A, \widetilde{L}_B)$, hence $[\widetilde{S}^-, \widetilde{S}^+] \supseteq [S^-, S^+]$. □

Offline compilation tabulates the three values $\big(U(r_j), L(r_j), q_{\mathrm{lo}}(r_j)\big)$ at 12 points per decade from $r = 0.1 \min(p_1, 0.005 \text{ bar})$ to $10^4 P_{\max}$ (median 192 points, a few kilobytes per record). Solves run single-threaded, one process per pair (median 9.9 s per pair). Given $(P, y)$, IsoCert bisects on $\ln(x/(1-x)) \in [-30, 30]$ to a bracket width of $10^{-3}$ using $U_A, \widetilde{L}_A, U_B, \widetilde{L}_B$ and reports each bracket's outer end. A root outside the search range makes that end infinite (a degenerate pair). Bisection needs no monotonicity of computed values because each returned end verifies one inequality between certified envelopes, which places $x^\pm$ on the correct side by Lemma 3. Across 5,200 queries on 208 pairs (Section VI-H), interval lookup takes a median of 0.15 ms per query (8118 times faster than direct LP evaluation). Compilation breaks even after 8 queries per pair and adds 0.014 median $\ln S$ width.

For threshold queries $S \geq \tau$ (Fig. 1(c)), IsoCert decides YES, NO, or UNDETERMINED via a direct *threshold predicate*. With

$$x_\tau = \frac{\tau y}{1 - y + \tau y}, \qquad r_A = \frac{Py}{x_\tau}, \qquad r_B = \frac{P(1-y)}{1 - x_\tau}; \quad (8)$$

the decision is YES iff $L_A(r_A) \geq U_B(r_B)$, NO iff $U_A(r_A) < L_B(r_B)$, and UNDETERMINED otherwise (Lemma 3). A threshold query needs four envelope values and no root solve. Online, the threshold predicate uses four table lookups (median 25 $\mu$s) or at most four direct LPs (median 4 LPs, 61 ms).

Alongside every YES or NO decision, IsoCert reports the *critical Henry cap* $H^\star = \kappa^\star K_{\min}$, the supremum of caps where the decision is certified (Section IV-B). Because certification ($S^-(\kappa) \geq \tau$ for YES, or $S^+(\kappa) < \tau$ for NO) is nonincreasing in $\kappa$, IsoCert computes $\kappa^\star$ by bisecting $\ln \kappa$ up to $\kappa = 10^5$. Since tables are compiled at one cap, bisection evaluates the threshold predicate by direct LPs at up to ten caps (two LPs per cap; median 0.4 s per decision).

### D. Acquisition Advisor

When a query is UNDETERMINED or its certified width $W = \ln S^+ - \ln S^-$ exceeds a target, IsoCert recommends the next measurement. By Theorems 3 and 4, interior points cannot remove the horizon bound of a Henry-like table and leave the hidden-capacity term unchanged. For curved records, interior points can still narrow the band. IsoCert instead uses *one-step lookahead*, a myopic heuristic motivated by the laws and not derived as optimal. Across 25 log-spaced candidate pressures in $[10^{-4}, 100]$ bar per gas $g \in \{A, B\}$, lookahead posits the band midpoint $\hat{q}_g(p) = \frac{1}{2}(q_{\mathrm{lo},g}(p) + q_{\mathrm{hi},g}(p))$ at tolerance $e_g$, recomputes $\widehat{W}(g, p) = \ln \widehat{S}^+ - \ln \widehat{S}^-$ by direct LPs, and picks $\arg\min \widehat{W}$. Section VI-G compares lookahead with a closed-form *horizon rule* sampling the gas with active IAST pressures $r_g^\pm$ furthest above $P_{\max,g}$, at $\min(r_g^-, r_g^+) \geq p_1$, until both gases are queried in-window.

## VI. Experiments

We evaluate IsoCert on 13,183 pure-component isotherms and 1,930 binary gas pairs from the NIST/ARPA-E Isotherm Database (ISODB) [1].

### A. Experimental Setup

**Data and Ingestion Protocol.** We retrieve 16,981 single-component isotherms from the public ISODB API [1] (2026-11-09) across the 14 gases in Table I at 250–400 K. Keeping records with pressure in bar, loading in mmol/g, and $n \geq 6$ positive pressures yields 13,183 records from 1,479 publications and 2,493 adsorbents (median 16 points). Records with $n > 40$ (11.8%) are thinned to 40 points by index.

**Tolerance, Admission, and Calibration.** Under full-scale tolerance $e_i = \varepsilon \max_j q_j$ [21], [22], we admit records with

TABLE I
ACROSS 13,183 ISODB RECORDS AND 14 GASES, $\mathcal{C}_{\mathrm{L}}$ ADMITS 68.5% AT $\varepsilon^\star \le 3\%$ (MEDIAN 1.54%). COLUMNS LIST RECORD COUNT, MEDIAN MISFIT FLOOR $\varepsilon^\star$ (%), AND ADMITTED SHARE (%).

| Gas | isoth. | $\varepsilon^\star$ | adm. | Gas | isoth. | $\varepsilon^\star$ | adm. |
|---|---|---|---|---|---|---|---|
| $CO_2$ | 5,776 | 1.43 | 69.4 | $C_3H_6$ | 318 | 1.94 | 63.2 |
| $CH_4$ | 2,716 | 1.51 | 72.9 | CO | 210 | 1.25 | 77.6 |
| $N_2$ | 1,698 | 1.97 | 61.7 | $C_2H_2$ | 194 | 0.94 | 85.1 |
| $H_2$ | 572 | 2.29 | 56.5 | $O_2$ | 191 | 1.12 | 76.4 |
| $C_2H_6$ | 423 | 1.79 | 65.7 | Ar | 106 | 1.32 | 75.5 |
| $C_3H_8$ | 407 | 3.31 | 47.4 | Xe | 94 | 2.17 | 57.4 |
| $C_2H_4$ | 394 | 0.95 | 81.7 | Kr | 84 | 1.32 | 88.1 |
| All 13,183 records | | | | median $\varepsilon^\star$ 1.54 | | | adm. 68.5 |

$\varepsilon^\star \le 3\%$ (Section V-A) and set $\varepsilon = \max(m\varepsilon^\star, 1\%)$ and $H = \kappa K_{\min}$ (default $\kappa = 10$, sensitivity $\kappa \in \{100, 1000\}$). We calibrate the single tuned hyperparameter $m$ on the held-out split (Section VI-C). From 3,051 records with $n \ge 12$ and $P_{\max}/p_1 \ge 30$, keeping points at $p \le P_{\max}/10$ ($\ge 6$ kept, $\ge 2$ hidden) yields 1,542 admitted prefixes with 32,832 hidden points. Because $P_{\mathrm{cut}}$ is the largest retained pressure, $\rho = r/P_{\mathrm{cut}}$ can exceed 10 (3,849 points); Table II bins the 28,983 points with $\rho \le 10$. Record-ID hashing splits these prefixes into 471 calibration (30%) and 1,071 test (70%) records. On the 280 calibration records with $P_{\max} \le 10$ bar (5,704 hidden points), the smallest $m \in \{1.25, 1.5, 2, 3\}$ with $\ge 95\%$ coverage fixes $m = 3$ (97.1% coverage, versus 92.0%, 87.6%, and 82.6% at $m = 2, 1.5, 1.25$).

**Baselines and Implementation.** Under the same $(T, \varepsilon, H)$, we compare $\mathcal{C}_{\mathrm{L}}$ against five baselines. Baseline (i), the *best-fit model*, takes the lowest-RMSE nonlinear least-squares fit (3–12 multi-starts [5], [6]) among Langmuir [9], dual-site Langmuir, Sips [10], [33], Tóth [11], UNILAN [12], and Freundlich. Baseline (ii), the uncertified *ensemble (of well-fitting families)*, reports $[\min, \max]$ over families (or pair combinations) with maximum residual $\le e_i$. Baseline (iii), the *resampling baseline* [7], reports $[\min, \max]$ over 121 pairs of 11 uniform-noise replicates in $[-e_i, e_i]$ via linear interpolation [5] with a *linear* or *flat* tail above $P_{\max}$. Baseline (iv) is the *hinge class (shape only)* $\mathcal{C}_{\mathrm{h}}(T, e, H)$ (Lemma 1). Baseline (v), the *dual-site parameter set* $\mathcal{C}_2$ (a computed, uncertified inner approximation, empty when no pair fits; median 0.16 s per pair), enumerates dual-site Langmuir isotherms within $(e_i, H)$. A logarithmic grid spans $10^{-4}/P_{\max}$ to $10^4/p_1$ at 10 affinities per decade plus $b = 0$, with 40 first-capacity values and the exact feasible second-capacity interval. A second level re-grids the feasible affinity bounding box at $44 \times 44$ with 56 capacities and takes the union. Refining to 16 per decade, 64 capacities, and $72 \times 72 \times 96$ changes $\ln S$ width by median 1.1% (90th percentile 4.4%, maximum 30.1%) on 296 pairs (0.4 s vs. 0.10 s per pair). The ensemble is uncapped (Sips and Freundlich have infinite Henry constants), and a record with no admitted family or an empty $\mathcal{C}_2$ counts as uncovered. The hinge class $\mathcal{C}_{\mathrm{h}}$ uses the same $e_i$ and $\kappa$ times its own $K_{\min}$. Runs use Python 3.12 and SciPy 1.18 HiGHS on a shared 32-core CPU server (no GPU; relative timings); code, record IDs, and results are in the supplement. Every reported $\mathcal{C}_{\mathrm{L}}$ and $\mathcal{C}_{\mathrm{h}}$ bound carries the continuum certificate of Lemma 4.

### B. Q1: Do Site-Mixture Classes Describe Real Data?

At the 3% gate, the Langmuir-site class $\mathcal{C}_{\mathrm{L}}$ admits 68.5% (9,033 of 13,183) of parsed ISODB records with median misfit floor $\varepsilon^\star = 1.54\%$ (Table I). Admission is 57.8% at 2% and 79.6% at 5%, while $\mathcal{C}_{\mathrm{h}}$ admits 76.6% (median 0.86%); records above the 3% gate are rejected.

### C. Q2: Do Certified Bands Hold on Hidden Real Measurements, and What Do Models Do There?

For $P_{\max} \le 10$ bar (607 test records, 12,402 hidden points above $P_{\mathrm{cut}}$), $[q_{\mathrm{lo}}(r), q_{\mathrm{hi}}(r)]$ under $\mathcal{C}_{\mathrm{L}}$ covers 97.0% of points up to extrapolation ratio $\rho = r/P_{\mathrm{cut}} = 10$ (95% record-level bootstrap interval 95.5%–98.3%; Table II, Fig. 1(a)–(b)). Resampling publications gives a 95% interval 95.3%–98.5% (235 publications), and coverage on the 187 test records from publications with no calibration record is 97.3%. Coverage is evaluated within the tolerance-widened band, which contains all admitted in-class fits respecting the cap. By contrast, the best-fit model stays within tolerance on 19.6% of points overall and 6.9% at $\rho \in (5, 10]$ (interval 5.1%–8.9%), where its relative error has median 16.9% and 90th percentile 65.8%. Single families cover 2.6%–6.3% in that bin, while the uncertified six-family ensemble covers 63.0% overall (interval 58.7%–67.3%). The uncertified dual-site parameter set $\mathcal{C}_2$ covers 95.3% of hidden points (94.5% in the far bin with median ratio 3.61, vs. 97.0% and 3.68 for $\mathcal{C}_{\mathrm{L}}$; Table II). This parameter set is empty on 1.6% of records. Calibrated on its own, $\mathcal{C}_2$ also selects $m = 3$ (95.1% calibration coverage). Keeping all feasible parameters of one family thus recovers most coverage lost by point fits and ensembles, while $\mathcal{C}_{\mathrm{L}}$ adds the remaining points and the continuum certificate. The Langmuir kernel ($\mathcal{C}_{\mathrm{L}}$) narrows the far-bin median ratio $q_{\mathrm{hi}}/q_{\mathrm{lo}}$ by 1.9× over $\mathcal{C}_{\mathrm{h}}$, from 6.85 to 3.68.

The validity boundary of $\mathcal{C}_{\mathrm{L}}$ appears at $P_{\max} > 10$ bar (464 records, 7,512 hidden points; Table II). Coverage drops to 84.2% with 15.1% of hidden points below $q_{\mathrm{lo}}$, versus 98.6% for $\mathcal{C}_{\mathrm{h}}$ and 49.8% for the ensemble. A plausible explanation is that surface-excess loading flattens faster than absolute loading at high pressure; we have not stratified the misses to test it. Within $P_{\max} \le 10$ bar, coverage is 97.0% instead of 100% because $\mathcal{C}_{\mathrm{L}}$ is a physical model.

### D. Q3: Is the Guarantee Exact When the Class Holds?

When ground-truth isotherms lie in $\mathcal{C}_{\mathrm{L}}$ and satisfy the Henry cap, $[S^-, S^+]$ covers the true $\ln S$ in 100.0% of 670 in-cap instances at $\kappa = 10$ (median width 0.70; Table III). With zero misses, the one-sided 95% upper bound on the miss rate across the 213 distinct source pairs is 1.4% (0.45% across all 670 instances). Coverage remains 100.0% of 687 at $\kappa = 100$ (width 1.00) and 100.0% of 698 at $\kappa = 1000$ (width 1.29). Crossing 360 real pairs with five truth families (Lemma 1) yields 1,800 tasks; keeping fits within 5% full-scale error resampled with $\pm 0.5\%$ noise leaves 990 instances. Excluding the true family,

TABLE II

ON HELD-OUT EXTRAPOLATION ($m = 3$) FOR $P_{\max} \leq 10$ BAR (607 TEST RECORDS, 12,402 HIDDEN POINTS), $\mathcal{C}_{\mathrm{L}}$ COVERS 97.0% WITH BANDS 1.9× TIGHTER AT $\rho \in (5, 10]$ THAN $\mathcal{C}_{\mathrm{h}}$, WHILE THE DUAL-SITE PARAMETER SET $\mathcal{C}_2$ COVERS 95.3%. COLUMNS LIST COVERAGE (%) AND MEDIAN WIDTH $q_{\mathrm{hi}}/q_{\mathrm{lo}}$ BY EXTRAPOLATION RATIO $\rho = r/P_{\mathrm{cut}}$ (BLUE BOLD, OURS; RED, COVERAGE $< 90\%$).

| | $\rho \in (1, 2]$ | | $\rho \in (2, 5]$ | | $\rho \in (5, 10]$ | | all $\rho \in (1, 10]$ | |
|---|---|---|---|---|---|---|---|---|
| Method | cov. | width | cov. | width | cov. | width | cov. | width |
| Best-fit model (point) | 62.3 | – | 19.5 | – | 6.9 | – | 19.6 | – |
| Six-family ensemble | 80.1 | 1.05 | 61.6 | 1.20 | 58.8 | 1.57 | 63.0 | 1.24 |
| Dual-site parameter set | 97.4 | 1.27 | 95.6 | 1.97 | 94.5 | 3.61 | 95.3 | 2.43 |
| Hinge class (shape only) | 99.0 | 1.51 | 98.6 | 3.11 | 99.1 | 6.85 | 98.9 | 4.29 |
| **Langmuir class (ours)** | **98.3** | **1.27** | **97.3** | **2.00** | **96.5** | **3.68** | **97.0** | **2.51** |
| *Outside the validity range: records with $P_{\max} > 10$ bar (464 isotherms, 7,512 hidden points)* | | | | | | | | |
| Dual-site parameter set | 95.2 | 1.19 | 80.2 | 1.72 | 78.9 | 3.04 | 82.0 | 2.05 |
| Hinge class (shape only) | 98.6 | 1.43 | 99.0 | 2.84 | 98.4 | 6.00 | 98.6 | 3.53 |
| Langmuir class | 96.0 | 1.20 | 81.8 | 1.76 | 82.0 | 3.13 | 84.2 | 2.14 |

TABLE III

ACROSS 990 KNOWN-TRUTH IAST INSTANCES ($\kappa = 10$, $P = 1$ BAR), ISOCERT ACHIEVES 100.0% COVERAGE OF $\ln S$ WHEN THE TRUTH SATISFIES THE HENRY CAP (LAST COLUMN, IN-CAP COUNT IN PARENTHESES), AGAINST 98.4% FOR THE COMPUTED DUAL-SITE PARAMETER SET ON THE SAME CAP-RESPECTING INSTANCES. SIPS† HAS A FINITE HENRY CONSTANT ONLY FOR EXPONENT ONE (0 INSTANCES; BLUE BOLD, OURS; RED, COVERAGE $< 90\%$).

| | | Share of instances whose interval contains the true $\ln S$ (%) | | | | | | |
|---|---|---|---|---|---|---|---|---|
| True family | $n$ | Ensemble (truth out) | Ensemble (truth in) | Resampling (linear tail) | Dual-site param. set | Hinge class | Langmuir class (all) | Langmuir class (cap holds) |
| Langmuir | 154 | 50.0 | 68.2 | 80.5 | 100.0 | 100.0 | 100.0 | **100.0** (154) |
| Dual-site Langmuir | 225 | 29.8 | 59.1 | 61.8 | 83.1 | 88.0 | 84.0 | **100.0** (166) |
| Tóth | 210 | 83.8 | 87.1 | 69.0 | 90.5 | 98.6 | 98.1 | **100.0** (164) |
| UNILAN | 190 | 83.2 | 86.8 | 75.3 | 95.3 | 98.9 | 98.9 | **100.0** (186) |
| Sips† | 211 | 48.3 | 81.5 | 62.1 | 86.7 | 97.6 | 96.7 | – (0) |
| All | 990 | 58.6 | 76.6 | 68.9 | 90.4 | 96.3 | 95.1 | **100.0** (670) |
| Median width of $\ln S$ | | 0.09 | 0.11 | 0.30 | 0.54 | 1.55 | **0.70** | |

the uncertified ensemble covers 58.6% and the best fit has median $|\Delta \ln S|$ error 0.033 (90th percentile 0.48). Resampling covers 68.9% (linear tail) and 46.0% (flat tail, median width 0.26), while $\mathcal{C}_2$ achieves 90.4% coverage overall and 98.4% when the cap holds (11 misses). Across 2,680 in-cap threshold queries ($\tau$ at 1.25× or 2× from true $S$), ISOCERT issues 1,592 certified YES/NO decisions (38.4% at 1.25×, 80.4% at 2×) with 0 wrong. On the same queries, $\mathcal{C}_2$ issues 1,783 uncertified decisions with 1 wrong, and the leave-one-out best fit errs on 6.3% and 2.2%. Over all 990 instances including cap-violating truths (Sips, 0 in-cap instances), $\mathcal{C}_{\mathrm{L}}$ coverage is 95.1%, 96.6%, and 97.0% at $\kappa = 10, 100, 1000$. When the truth violates $\kappa = 10$, 4.8% of certified decisions fail because the cap is an explicit physical assumption.

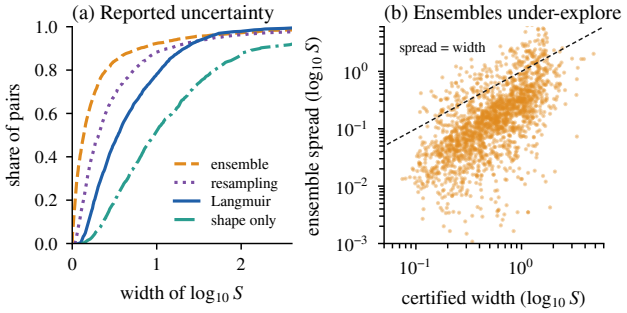



Fig. 3. Across 1,930 real pairs ($P = 1$ bar), uncertified fits under-report selectivity ambiguity. (a) Distributions of $\log_{10} S$ width for ensemble (orange), linear-tail resampling (purple), $\mathcal{C}_{\mathrm{h}}$ (teal), and $\mathcal{C}_{\mathrm{L}}$ (blue). (b) Ensemble spread versus certified width (log-log, dashed equality line, $\rho_s = 0.59$).

### *E. Q4: What Do Real Pairs Certify, and What Does Current Practice Report?*

Across 1,930 real pairs at $P = 1$ bar and $\kappa = 10$, $\mathcal{C}_{\mathrm{L}}$ certifies a median $\ln S$ width of 1.25, a factor of 3.5 between $S^-$ and $S^+$ (IQR 0.69–2.15; Table IV, Fig. 3(a), Fig. 1(c)). From 3,799 matched adsorbent/publication/±1 K pairs, 1,983 are admitted, 1 has an empty class at the working tolerance, and 52 reach $|\ln(x/(1-x))| = 30$, leaving 1,930. The uncertified ensemble reports a median $\ln S$ spread of 0.30 (factor 1.36) and spans only 26% of the certified width. Resampling spans 60% (linear tail) and 50% (flat tail, median 0.53) and lies inside $[S^-, S^+]$ on 51% of pairs, while $\mathcal{C}_{\mathrm{h}}$ is a median 1.57× wider per pair (factor 9.4). Ensemble spread has Spearman correlation $\rho_s = 0.21$ with pair fit RMSE and $\rho_s = 0.59$ with certified width (Fig. 3(b)). All 20,988 of 20,988 in-class, in-cap ensemble combinations lie inside $[S^-, S^+]$, whereas the minimum-RMSE fit falls outside in 14.6% of pairs. The

TABLE IV

ACROSS 1,930 REAL PAIRS AND 12 GAS COMBINATIONS ($P = 1$ BAR, $\kappa = 10$), ENSEMBLE AND RESAMPLING SPREADS CAPTURE ONLY 26% AND 60% OF THE CERTIFIED $\ln S$ WIDTH, WHILE THE DUAL-SITE PARAMETER SET $\mathcal{C}_2$ (UNCERTIFIED INNER BOUND) SPANS 96%. COLUMNS LIST PAIRS, MEDIAN SPREADS (INCLUDING $\mathcal{C}_2$) AND CERTIFIED WIDTHS OF $\ln S$, SPEARMAN $\rho_s$ OF ENSEMBLE SPREAD WITH FIT RMSE, AND YES/NO SHARES CERTIFIED AT $2\times$ FROM THE BEST FIT (BLUE BOLD, OURS).

| | | Uncertified spread of $\ln S$ | | | | Certified width of $\ln S$ | | Certified at $2\times$ (%) | |
|---|---|---|---|---|---|---|---|---|---|
| Gas pair ($y_A$) | pairs | ensemble | resampl. | dual-site set | $\rho_s$(fit) | hinge | **Langmuir** | hinge | **Langmuir** |
| $CO_2/CH_4$ (0.50) | 699 | 0.33 | 0.76 | 1.12 | 0.21 | 2.24 | **1.30** | 52/18 | **64/49** |
| $CO_2/N_2$ (0.15) | 355 | 0.45 | 1.04 | 1.53 | 0.24 | 3.28 | **1.72** | 57/9 | **66/29** |
| $CH_4/N_2$ (0.50) | 314 | 0.33 | 0.85 | 1.16 | 0.16 | 2.01 | **1.22** | 48/25 | **58/51** |
| $C_2H_4/C_2H_6$ (0.50) | 138 | 0.11 | 0.31 | 0.57 | −0.13 | 1.06 | **0.73** | 61/59 | **75/75** |
| $C_2H_6/CH_4$ (0.50) | 79 | 0.32 | 0.61 | 1.14 | 0.19 | 3.34 | **1.25** | 58/19 | **75/48** |
| $O_2/N_2$ (0.21) | 74 | 0.19 | 0.44 | 0.74 | 0.22 | 1.33 | **0.80** | 53/61 | **76/68** |
| $CO_2$/CO (0.50) | 59 | 0.40 | 0.48 | 1.37 | 0.09 | 2.97 | **1.48** | 47/17 | **54/51** |
| $C_3H_6/C_3H_8$ (0.50) | 58 | 0.24 | 0.27 | 0.90 | −0.22 | 2.04 | **1.36** | 36/29 | **66/50** |
| $C_2H_2/C_2H_4$ (0.50) | 51 | 0.09 | 0.26 | 0.53 | 0.37 | 1.00 | **0.61** | 88/61 | **96/88** |
| $C_2H_2/CO_2$ (0.50) | 47 | 0.15 | 0.37 | 0.47 | 0.42 | 0.91 | **0.51** | 96/57 | **96/94** |
| $CO_2/H_2$ (0.20) | 34 | 0.61 | 0.83 | 2.32 | 0.27 | 3.79 | **2.37** | 15/24 | **29/26** |
| Xe/Kr (0.20) | 22 | 0.13 | 0.82 | 0.78 | −0.16 | 1.74 | **0.81** | 55/36 | **73/77** |
| All pairs | 1,930 | 0.30 | 0.72 | 1.10 | 0.21 | 2.24 | **1.25** | 54/25 | **66/50** |

uncertified dual-site parameter set $\mathcal{C}_2$ (Table IV) has median width 1.10 and spans a median 96% of the certified width (lower quartile 90%, 10th percentile 77%). It is non-empty for 97.3% of pairs and lies inside $[S^-, S^+]$ for 100.0%. Because few-site members attain $S^-$ and $S^+$ (Theorems 1–2), two-site models realize this factor-3.5 ambiguity, while $[S^-, S^+]$ exceeds $\mathcal{C}_2$ by a median of 4.0% to bracket the range. Median $\ln S$ width grows to 1.71 and 2.17 at $\kappa = 100$ and 1000 and narrows to 0.90 at $m = 2$. Median width is 2.29 at $P = 0.1$ bar and 2.21 at $P = 10$ bar. Keeping all points on the 295 pairs with a thinned record changes the width by a median factor 1.00 (quartiles 1.00–1.02).

Existing records still certify most screening queries once $\tau$ is separated from the point estimate (Table IV). For $2\times$ thresholds from the best fit, raising $\kappa$ across $10, 100, 1000$ shifts certified YES shares across 66%, 57%, 51% and NO shares across 50%, 37%, 29%. Bisecting on $\ln \kappa$ up to $10^5$ over 500 sampled pairs, 85%, 75%, and 62% of the 350 YES decisions certified at $\kappa = 10$ remain certified at $\kappa = 100, 1000, 10^5$. Of the 255 NO decisions, 73%, 58%, and 36% remain, with median critical multiplier $3.9 \times 10^3$; decisions with $\kappa^\star$ near 10 depend on the Henry cap.

**Measured Mixtures (Outside the Guarantee).** We match 272 measured ISODB mixtures to admitted pure-gas records from the same publication, adsorbent, and temperature (2,512 mixture points, 109 publications, 33 gas pairs). Without added mixture tolerance, $[S^-, S^+]$ contains the measured selectivity $(n_A/n_B)/(y_A/y_B)$ on 60% of points at $\kappa = 10$ (publication-bootstrap 52%–67%) and 75% at $\kappa = 1000$. By contrast, $\mathcal{C}_\mathrm{h}$ covers 73%, $\mathcal{C}_2$ covers 49%, and the ensemble covers 24%, while the best-fit IAST prediction has median $\ln S$ error 0.39. The 25% of points outside $[S^-, S^+]$ at $\kappa = 1000$ cannot be reproduced by IAST with any member pair (median best-fit error 0.93, versus 0.18 inside). This test does not identify whether IAST, a pure-gas class, the tolerance, the cap, or the mixture measurement fails; real-mixture non-ideality [23] lies outside our binary IAST guarantee.

### F. Q5: Do the Two Laws Hold?

The analytical witnesses of Theorems 3 and 4 match the semi-infinite LP envelopes and govern real records (Fig. 4). Across 300 synthetic Henry-like configurations ($\varepsilon \in \{0.25, 0.5, 1, 2, 4\}\%$, $n \in \{6, 12, 24, 48\}$, three cap regimes, $\rho \in \{3, 10, 30, 100, 300\}$), the LP lower envelope $L(r)$ divided by the Theorem 3 witness lies between 0.9845 and 1 (median 1.0000; Fig. 4(a)). Increasing $n$ from 6 to 48 alters $L(r)$ by at most 1.57%, whereas the $\mathcal{C}_\mathrm{h}$ lower envelope grows at most logarithmically above $P_\mathrm{max}$. Across 857 real records (the admitted, feasible members of a random sample of 1,200 parsed records), the slope $\mathrm{d}U(P_\mathrm{max}/2; H)/\mathrm{d}\ln H$ between $\kappa = 10^4$ and $10^5$ divided by the LP hidden capacity $c_0$ has median 0.9920 (IQR 0.924–0.9999, 10th percentile 0.84; Fig. 4(b)). The Theorem 3 scaling $\varepsilon(\rho - 1)$ also collapses the 32,308 hidden points with $\rho > 1.2$ from all 1,542 prefixes of Section VI-C (both splits and pressure regimes, $m = 3$; Fig. 4(c)). A descriptive regression of $\log_{10}(q_\mathrm{hi}/q_\mathrm{lo} - 1)$ on $\log_{10}(\varepsilon(\rho - 1))$ gives slope 0.81 (prefactor 7.3), $R^2 = 0.91$, and Spearman 0.94.

### G. Q6: Which Measurement Makes a Query Certain?

One-step lookahead reaches target $\ln S$ width $\leq 0.25$ ($\leq 8$ measurements) on 68% of 100 instances (median 2), versus 67% (median 4) for round-robin, 51% for random, and 7% for densify. Steps count measurements acquired before reaching the target (unreached runs count as 9 in means). From 540 dual-site Langmuir and Tóth tasks ($p \leq 1$ bar, $\pm 0.5\%$ noise, $\varepsilon = 1\%$, $\kappa = 10$), excluding 224 poor fits, 117 with $< 6$ points, 51 cap violations, and 48 already-certified tasks leaves 100 instances (median initial width 0.46). All policies choose from the same 25 log-spaced candidate pressures per gas in $[10^{-4}, 100]$ bar. An observation is the true value plus uniform

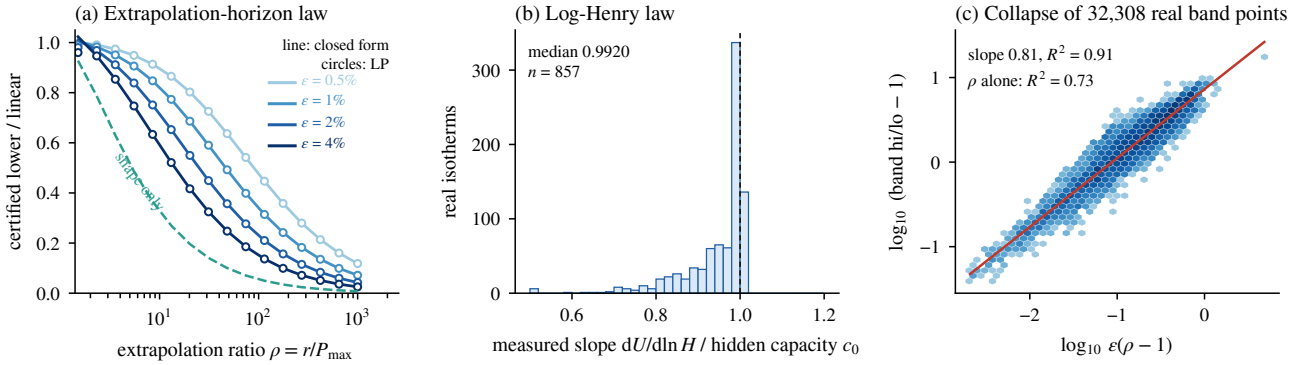



Fig. 4. Analytical witnesses (Theorems 3–4) match LP envelopes and collapse extrapolation widths. (a) $L(r)/(Kr)$ versus $\rho$ for the Theorem 3 witness (blue lines), LP (circles), and $\mathcal{C}_{\mathrm{h}}$ at 1% (dashed teal). (b) Histogram over 857 records of $(\mathrm{d}U/\mathrm{d}\ln H)/c_0$ (median 0.9920). (c) Collapse against $\varepsilon(\rho-1)$ over 32,308 points ($R^2 = 0.91$).

noise of $\pm 0.5\%$ of full scale, and after each observation the class is rebuilt with the same tolerance and cap rule. Round-robin alternates gases every two steps and extends the window by a factor 3 upward (at most 100 bar) on even steps and downward (at least $10^{-4}$ bar) on odd steps. Random draws an unused candidate uniformly. Densify alternates gases and adds the geometric midpoint of the largest logarithmic gap inside the window. Paired, lookahead needs fewer measurements than round-robin on 61% of instances and more on 7% (mean saving 1.4, bootstrap 1.0–1.9). Lookahead first measures the weak gas above $P_{\max}$ in 84% of instances. Closed-form rules do not replace lookahead; the horizon rule stops after a mean of 2.7 measurements and reaches the target on 42% (variant 45%). Lookahead takes 109.6 s per instance against 2.5 s.

### H. Q7: Cost: Compile Once, Query Fast

Pre-compiling $(U, L, q_{\mathrm{lo}})$ once per record at 12 grid points per decade speeds up online interval queries by 8118× in median latency. Across 208 admitted $CO_2/N_2$ and $CO_2/CH_4$ pairs over 5 pressures (0.1–10 bar) and 5 compositions (0.05–0.8; 5,200 queries, median 192 grid points per pair), direct evaluation solves median 73 column-generation LPs in 1235 ms per query. Compiled lookup (Proposition 1) takes a median of 0.15 ms per query, contains $[S^-, S^+]$ in 100.0% of queries, and adds a median $\ln S$ width of 0.014. On 15,285 threshold queries, the compiled threshold predicate takes a median of 25 $\mu$s against 61 ms for the direct predicate (4 LPs). Compiled lookup keeps 100.0% of direct decisions and never contradicts one (0). Critical-cap search takes a median of 0.4 s per decision by direct LPs, and acquisition advice takes 109.6 s per instance. Across 400 off-grid synthetic few-site pairs with between-grid affinities (200 with a site beyond the column range) and tight Henry caps, 0 of 16,000 envelope and 0 of 1,039 interval checks fail. Likewise, 0 of 2,038 certified threshold decisions (thresholds within $10^{-3}$ of truth and interval ends) contradict the truth. In a solver ablation on 300 real pairs, dropping column generation widens the certified interval by a median of 2.1% (90th percentile 4.6%). The raw grid LP without the continuum certificate (an uncertified bound) is narrower than $[S^-, S^+]$ in 93% of pairs (median 0.26%). Compilation takes a median of 9.9 s per pair (about 12 CPU-hours across all 9,033 admitted records in parallel). Because compilation breaks even after 8 queries, direct evaluation is faster for single-shot queries.

## VII. Limitations and Conclusion

**Limitations and scope boundaries.** First, certain answers are relative to the IAST mixing rule [2] and stated class, tolerance, and Henry cap; real-mixture departures from IAST [23] lie outside scope. On measured mixtures, the certified interval contains 60% of points (75% at $\kappa = 1000$), bounding what an IAST-relative guarantee delivers. Because $U(r; H)$ diverges as $H \to \infty$ under tolerance slack (Theorem 4), the cap is substantive. When the cap fails, 4.8% of certified decisions err versus 0 of 1,592 in-cap, so we report the critical cap. Second, $\mathcal{C}_{\mathrm{L}}$ covers 97.0% of held-out points at $P_{\max} \le 10$ bar (versus 100.0% in-class truth coverage). Above 10 bar, coverage drops to 84.2% (15.1% below the band, versus 98.6% for $\mathcal{C}_{\mathrm{h}}$); a plausible, untested explanation is that excess adsorption flattens faster than absolute adsorption. Third, certified intervals have median selectivity factor 3.5 at $\kappa = 10$ (excluding 52 degenerate pairs), an intrinsic limit of noisy tables (Theorems 3–4). The computed dual-site parameter set is 4.0% narrower but misses 11 cap-respecting truths and carries no certificate. Beyond two components, monotonicity bounds each adsorbed fraction (upper envelope vs. lower for others), though selectivity bounds are not sharp. Finally, closed-form horizon rules reach the target on 42% of instances versus 68% for lookahead. Each continuum certificate holds up to double-precision rounding (certificate margin at most $1.9 \times 10^{-6}$).

**Conclusion.** We formulate certain-answer screening over site-mixture classes $(\mathcal{C}_{\mathrm{h}}, \mathcal{C}_{\mathrm{L}})$ and reduce $[S^-, S^+]$ to two $(n+2)$-site envelopes per record (Theorems 1–2). Two laws (Theorems 3–4) bound what finite tables certify on Henry-like records and under tolerance slack. IsoCert compiles sound envelope tables (Proposition 1) and answers interval queries in 0.15 ms (8118× faster than direct evaluation). Certified bands cover 97.0% of held-out points and 96.5% at 5–10× extrapolation, where best-fit models stay within tolerance on 6.9%.

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